TIME SERIES FORECASTING

Streaming Data Management and Time Series Analysis

SECTION 1: Exploratory Time Series Analysis (EDA)

Description of the Data

The dataset student_dataset.csv contains the hourly count of pedestrians recorded by a sensor installed on a public street in Australia. The time series spans from April 15, 2015, to February 29, 2020, for a total of 42,767 hourly observations. Among these, 41,328 observations are available and constitute the training set, while the remaining 1,439 values are missing and must be forecasted.

The target variable is a non-negative integer count ranging from 0 to 906. Time is recorded in the format yyyy-mm-dd / hh:mm:ss in UTC. The continuity of the series has been verified, confirming the absence of missing hourly timestamps within the training period.

Table 1 - Dataset Structure

Feature	Value
Training observation	41.328
Observations to be expected	1.439
Start date	2015-04-15 00:00:01 UTC
End of training	2019-12-31 23:00:01 UTC
Start of forecast	2020-01-01 00:00:01 UTC
End of forecast	2020-02-29 22:00:01 UTC
Frequency	Hourly
Gap in the series	Nobody
Leap year 2020	Yes – February 29 days

Descriptive Statistics

The first step of the exploratory analysis involved computing summary statistics to characterize the distribution of the target variable and inform subsequent modeling decisions.

Table 2 - Descriptive statistics of the Training set

Statistics	Value	Interpretation
Minimum	0	Night hours/holidays
1st Quartile	14,0	Low attendance
Median	69,0	Typical value
Mean	101,5	High from daytime peaks
3rd Quartile	159,0	Afternoon slot
Maximum	906,0	Exceptional peak
Standard Deviation	103,17	High variability
CV	101,6%	SD > mean
Skewness	1,144	Right-skewed distribution
Kurtosis	3,854	Heavy tails
Proportion of zeros	2,66%	Night hours with active sensor

A comparison between the mean (101.5) and the median (69.0) clearly indicates positive skewness, with a long right tail driven by high pedestrian counts during weekday afternoons.

The coefficient of variation (CV = 101.6%) exceeds 100%, meaning that the standard deviation is greater than the mean. This highlights the strong variability of the series, reflecting the contrast between near-zero nighttime values and peak daytime activity (up to 906 observations).

These characteristics motivated the application of a Box-Cox transformation prior to modeling, with the aim of stabilizing the variance and improving the reliability of estimation methods such as OLS, ARIMA, and UCM.

Visual Analysis of the Series

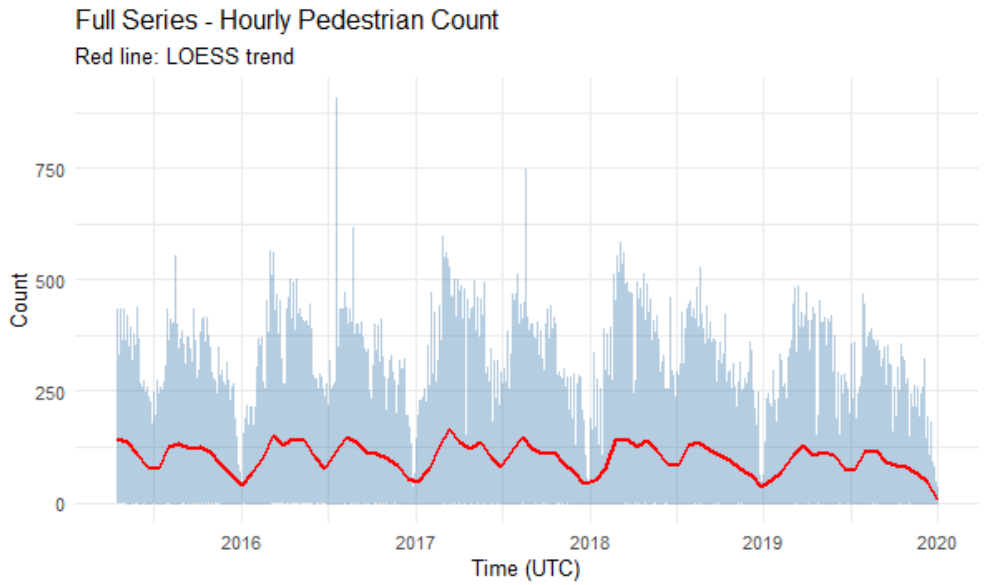


Figure 1- Complete series with non-parametric LOESS trend

The full time series highlights the dominant structure of the data: overlapping seasonal oscillations at multiple frequencies. The LOESS trend (span = 0.05) reveals a long-term oscillatory pattern associated with annual seasonality, with higher values during the Australian summer (January–February) and lower values in winter (June–August). No persistent linear trend is observed over time, an insight that is relevant for determining the integration order d in ARIMA and UCM models.

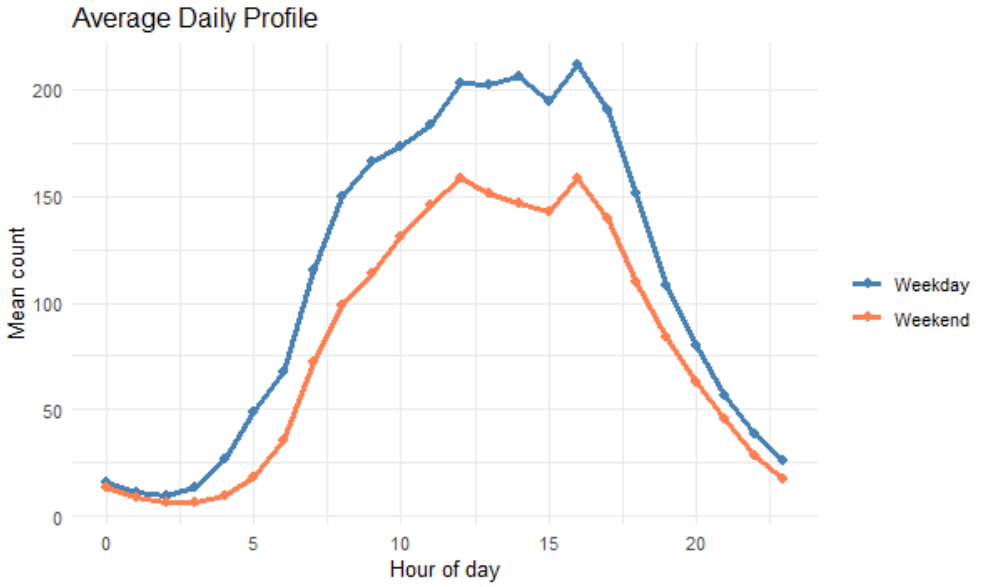


Figure 2 - Average hourly profile by type of day

The average hourly profile shows a clear unimodal pattern, with a peak between 2 PM and 4 PM. The absence of a bimodal structure typical of commuting patterns suggests that the sensor is located in a commercial or tourist area, where activity is concentrated during daytime hours.

A significant difference is observed between weekdays and weekends: the average hourly count is 110.4 during weekdays and 79.4 during weekends (–28%). This difference motivated the inclusion of a weekend dummy variable in all models.

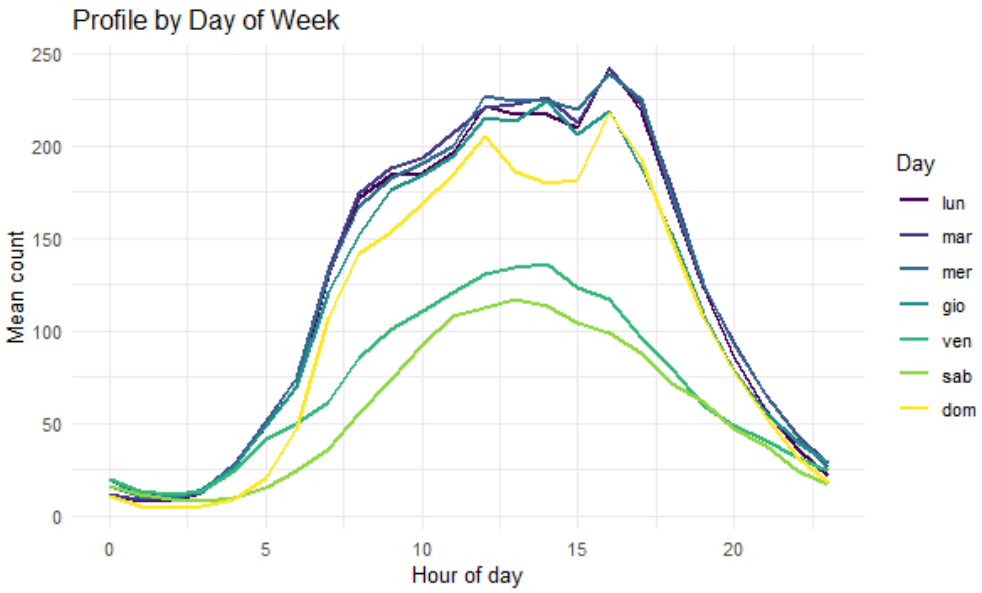


Figure 3 - Average hourly profile by day of the week

Monday, Tuesday, and Wednesday exhibit nearly identical profiles, with peaks exceeding 220 observations per hour. Thursday and Friday show slightly lower levels, while Saturday and Sunday display significantly reduced activity, with later start times and earlier declines. This pattern justifies the inclusion of a weekly seasonal component (s = 168 hours), in addition to daily seasonality (s = 24 hours).

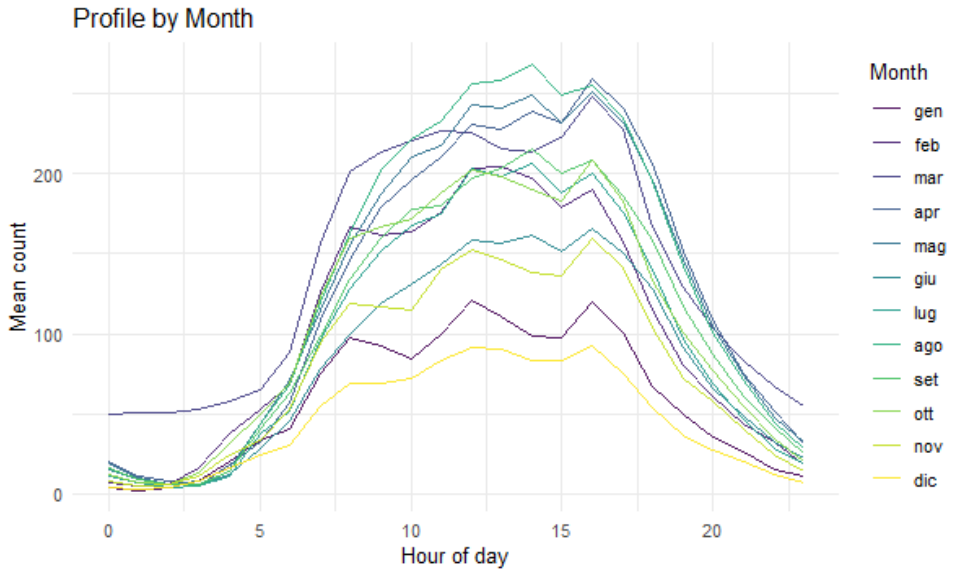


Figure 4 - Average hourly profile by month

The monthly profile confirms a clear annual seasonal pattern. January and February (Australian summer) show the highest peak values, exceeding 280 observations per hour, while June and July correspond to the lowest levels. December appears anomalously low compared to other summer months, likely due to the effect of Christmas holidays. Since the forecast period (January–February 2020) falls within a seasonal peak, this aspect is crucial for evaluating forecast accuracy.

Special Events Analysis

The impact of major Australian holidays was quantified by comparing their average hourly values with the reference weekday mean (110.4 observations per hour).

Table 3 - Quantified impact of Australian holidays

Event	Days	Hourly average	Vs. Average weekday
Christmas	24-26 dec	13,0	-88%
New Year's Eve	1 jan	27,3	-75%
Australia Day	26 jan	40,4	-63%
ANZAC Day	25 apr	133,0	-17%
Weekend	Sun-Sat	79,4	-28%

Christmas and New Year's Day show the largest reductions, with decreases of 88% and 75%, respectively. These events fall within the forecast period and must therefore be explicitly modeled.

Australia Day also has a strong negative effect (–63%), while ANZAC Day shows a more moderate reduction (–17%). Despite its smaller impact, ANZAC Day is included due to its statistical significance in the training data.

Seasonal Decomposition

To further investigate the internal structure of the series, a multiple STL decomposition was applied with seasonal periods of 24 and 168 hours.

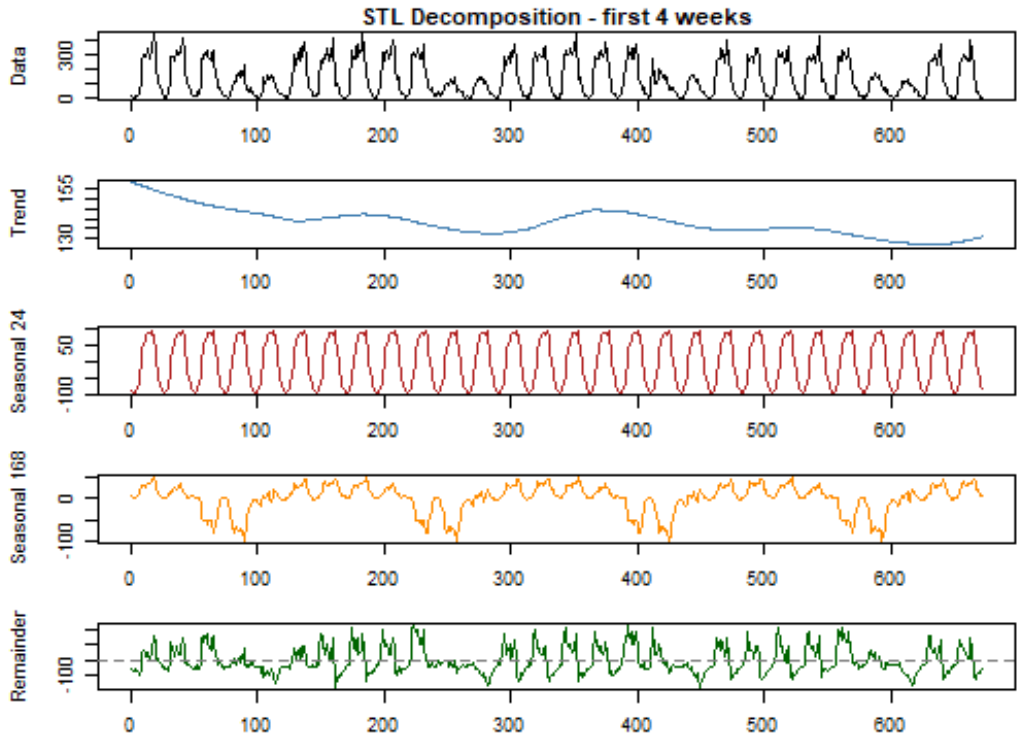


Figure 5 - STL Decomposition

The results confirm the presence of a strong daily seasonal component and a moderate weekly pattern, while the trend appears weak and slowly varying, without a clear linear drift.

These findings support the modeling choices adopted in the subsequent sections: the inclusion of both daily and weekly seasonal components, and the use of specifications without differencing (d=0) or stochastic slope.

Stationarity Analysis

Stationarity is a fundamental assumption for ARIMA models. Three statistical tests were applied to determine the appropriate order of integration.

Table 5 - Stationarity test.

Test	Null hypothesis	P – value	Conclusion
ADF (Augmented Dickey- Fuller)	Non-stationary series	< 0,01	Stationary → d=0 defensible
KPSS (level)	Stationary series	< 0,01	Non-stationary
KPSS (trend)	Trend-stationary	< 0,01	Stochastic trend

The ADF test rejects the null hypothesis of non-stationarity ($p < 0.01$), supporting the use of d=0. In contrast, the KPSS test rejects the null hypothesis of stationarity, indicating the presence of a stochastic trend.

This apparent contradiction is common in long seasonal time series: the ADF captures cyclical stationarity, while KPSS detects deviations from a constant level due to slow structural changes. Overall, the series can be interpreted as stationary with a stochastic trend and deterministic seasonal structure. Both d=0 and d=1 are theoretically justifiable; however, d=0 was selected to avoid the forecast anchoring problem observed when differencing is applied.

Autocorrelation Analysis

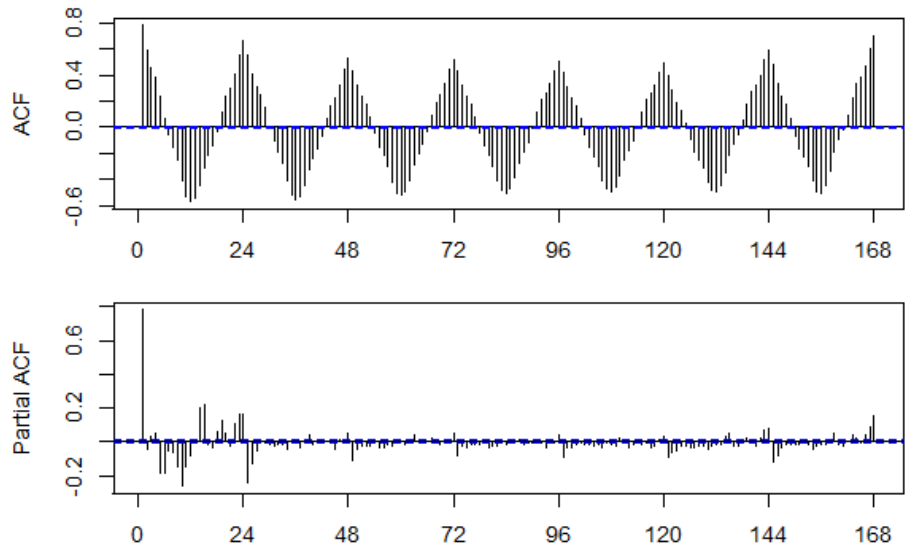


Figure 6 - ACF and PACF of the original series

The ACF shows strong and persistent peaks at multiples of 24, indicating pronounced daily seasonality and long-range dependence. The PACF rapidly decays after the first lags (approximately lag 5–10), suggesting a low-order autoregressive structure once seasonality is removed.

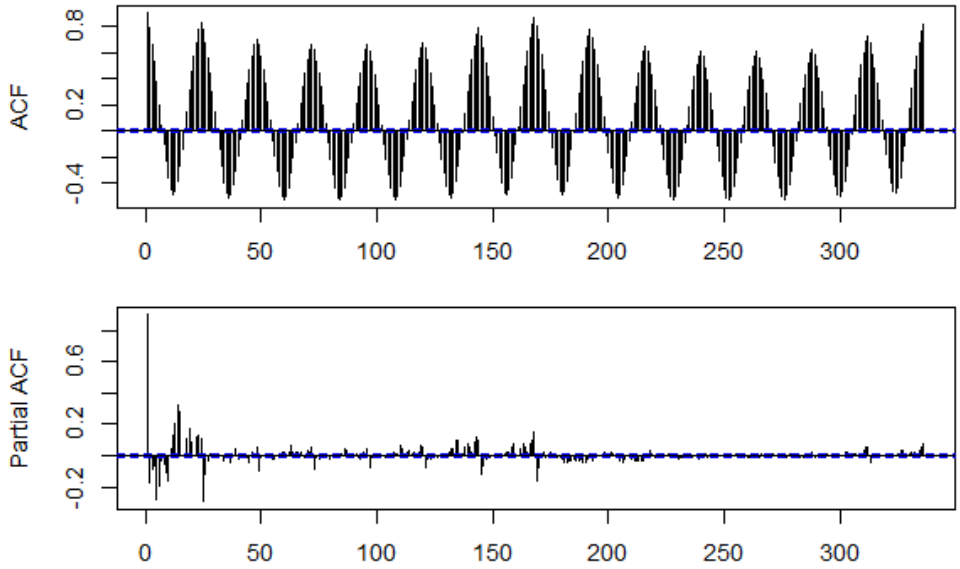


Figure 7 - ACF and PACF of STL residuals

The residual ACF still presents significant peaks at multiples of 24, confirming that seasonal structure is not entirely removed by decomposition alone. This supports the inclusion of an ARMA component in subsequent models. The PACF suggests an autoregressive order between 1 and 3.

EDA Conclusions and Modeling Implications

The exploratory analysis leads to the following key conclusions:

- Strong deterministic double seasonality (24h and 168h), requiring explicit modeling
- Positively skewed distribution with high variability, motivating a Box-Cox transformation
- Weak trend without stable linear drift, supporting d=0 in ARIMA and a Local Level in UCM
- Significant holiday effects, requiring the inclusion of dummy variables
- Low-order residual autocorrelation, suggesting limited ARMA terms
- Forecast period located in peak seasonal conditions (austral summer), requiring accurate modeling of seasonal levels

SECTION 2: ARIMA Model

Motivation and Modeling Strategy

ARIMA (AutoRegressive Integrated Moving Average) models represent a standard benchmark for time series forecasting. The extended ARIMAX formulation allows the inclusion of exogenous regressors, enabling a separation between:

- the **deterministic structure** of the series (modeled through regressors such as Fourier terms and holiday dummies),
- the **stochastic residual component** (modeled through ARMA dynamics).

The modeling strategy follows directly from the insights obtained in the EDA:

- A Box-Cox transformation is applied to stabilize variance and improve residual normality.
- Fourier regressors are used to capture multiple seasonalities (24h and 168h), avoiding the need for seasonal differencing.
- Dummy variables are introduced to model the impact of holidays.
- The integration order is fixed at d=0, based on both statistical evidence and forecasting considerations.

- The ARMA structure is selected automatically using auto.arima(), minimizing the AICc across candidate models.

Box-Cox Transformation

The Box-Cox transformation is applied prior to model estimation to address the strong skewness and heteroscedasticity identified in the EDA.

Let $y_{raw} = value + 0.5$, where the offset accounts for zero observations (2.66% of the dataset). The transformed series is defined as:

$$y_{bc} = \frac{y_{raw}^{\lambda} - 1}{\lambda}, \qquad \lambda = 0.2491$$

The parameter λ=0.2491, estimated using Guerrero's method, corresponds to a root-type transformation. This compresses extreme peaks while preserving meaningful differences between low and high values.

At the forecasting stage, predictions are back-transformed using the inverse Box-Cox function and subsequently clipped at zero to ensure non-negative values.

Fourier Regressors

Seasonality can be modeled either through SARIMA structures or through deterministic Fourier terms. In this context, Fourier regressors are preferred for three main reasons:

- **Multiple periodicity:** SARIMA models handle only one seasonal frequency, whereas the series exhibits both daily (24h) and weekly (168h) cycles.
- **Parsimony:** Fourier terms require fewer parameters compared to seasonal ARMA components.
- **Flexibility:** the number of harmonics K can be tuned using information criteria.

Fourier terms are defined as:

$$S_k(t) = \sin\left(\frac{2\pi kt}{s}\right), \qquad C_k(t) = \cos\left(\frac{2\pi kt}{s}\right)$$

Three seasonal components are modeled using Fourier regressors:

Table 6 - Fourier regressors included in the model

Frequency	Period (s)	Optimal K	Regressors	Captured Seasonality
Daily	24 hours	12	24	Intra-daily profile (peak at 2-4 PM)
Weekly	168 hours	6	12	Weekday – weekend cycle (-28% Saturday – Sunday)
Annual	8,766 hours	1	2	Seasonal modulation (austral summer vs winter)

The optimal number of harmonics was selected through grid search, minimizing the AICc.

K daily	K weekly	K annual	AICc	Notes
8	4	1	128.539,83	
10	4	1	128.446,31	
12	4	1	128.380,84	
8	6	1	128.317,91	
10	6	1	128.395,59	
12	6	1	128.158,73	Optimal
8-12	8	1-2	-	Error (regressor collinearity)
10	8	2	128.247,12	
12	8	2	128.308,34	

Table 7 - Grid search AICc for optimal K. Combinations with K_weekly=8 produced estimation errors.

The optimal configuration is K=(12,6,1), corresponding to daily, weekly, and annual components, respectively. Higher values for the weekly component (e.g., K=8) led to estimation issues due to collinearity.

Holiday Dummy Variables

Based on the EDA, five binary dummy variables were included to capture the effect of holidays:

Dummy	Days included	Historical impact	Training hours	Forecast hours
Christmas	24 - 26 dec	-88% vs weekday	360	0 (only training)
New_year	31 dec – 2 jan	-75% vs weekday	312	48 (Jan 1-2, 2020)
Australia_day	26 jan	-63% vs weekday	96	24 (Jan 26, 2020)
Anzac_day	25 apr	-17% vs weekday	120	0 (only training)
weekend	Sat and Sun	-28% vs weekday	11.808	Yes

Table 8 - Holiday dummy variables with impact quantified by the EDA.

Christmas and ANZAC Day do not fall within the forecast horizon but contribute to coefficient estimation during training. In contrast, New Year and Australia Day occur within the forecast period and are therefore critical for accurate predictions.

Choice of Integration Order (d = 0)

The decision to fix d=0 is both theoretically and empirically motivated.

In an initial unconstrained specification, auto.arima() selected a model with d=1, leading to severe underestimation of forecasts (mean ≈ 17 units/hour versus expected ≈ 100).

This issue arises because differencing transforms the model into a process anchored to the last observed value. Since the final observation corresponds to a seasonal minimum (December 31, late evening), the model fails to capture the transition to the higher seasonal level of January–February.

By setting d=0, the model preserves the level of the series, while seasonal patterns are captured through Fourier regressors. This allows the forecast to correctly reflect the transition from winter to summer conditions.

The choice is supported by:

- the ADF test, which confirms stationarity ($p < 0.01$),
- the practical requirement of avoiding the anchoring problem,
- the ability of Fourier terms to capture slow seasonal variation.

Model Estimation

The final model was estimated using auto.arima() with d=0, stepwise = TRUE, and approximation = TRUE to reduce computational cost. The selected model is:

$$ARIMA(5,0,3)$$

with the following structure:

$$y_{bc}(t) = \sum_{i=1}^{5} \phi_i y(t-i) + \sum_{j=1}^{3} \theta_j \varepsilon(t-j) + \beta X(t) + \varepsilon(t)$$

Parameter	Value
ARIMA Model	ARIMA(5,0,3)
Ar1	1,8192 (± 0.0039)
Ar2	–0,8869 (±0,0038)
Ma1	–2,0321 (±0,0066)
Ma2	1,1698 (±0,0121)
Ma3	–0,1268 (±0,0057)
S1 – 24 (Daily Fourier)	–2,6208
C1 -24 (Daily Fourier)	–3,3765
C1 – 168 (Weekly Fourier)	0,9397
New_year	–0,1585
Australia_day	0,1631
Anzac_day	0,4331
Weekend	0,1144
σ^2	1,307
AICc	128.381,60
BIC	128.795,70

Table 9 - Main coefficients of the ARIMA(5,0,3) model.

The model achieves:

- AICc=128,381.60
- BIC=128,795.70

Residual Diagnostics

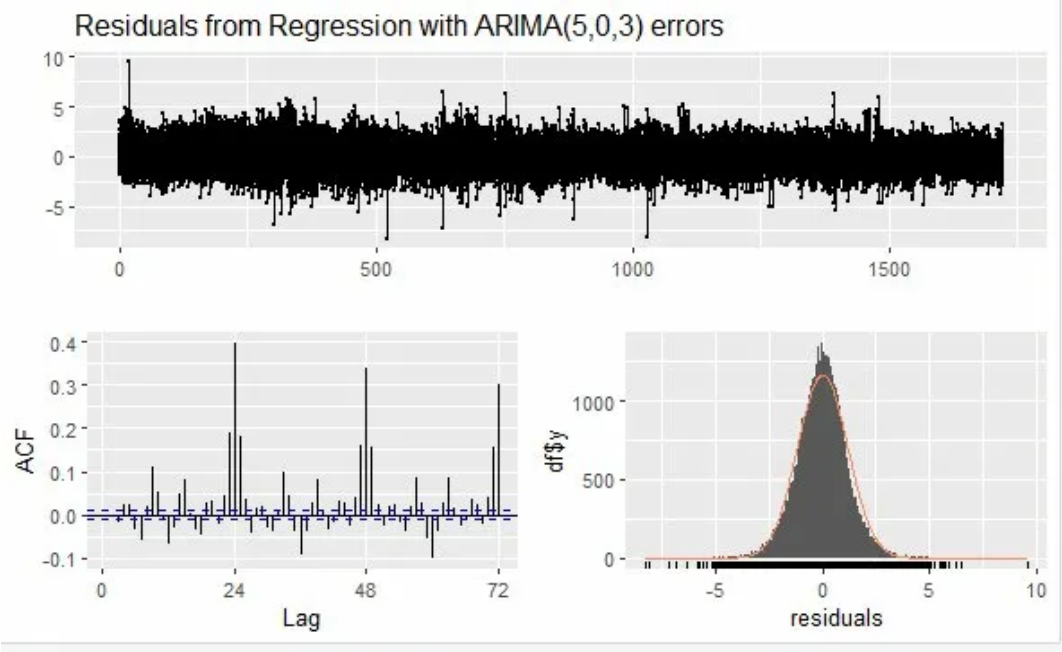


Figure 8 – Residual diagnostics

Figure 8

The residual series is centered around zero and shows approximately constant variance. The distribution is close to normal, although slightly heavy-tailed.

The ACF still exhibits small but significant spikes at multiples of 24, indicating residual seasonal dependence not fully captured by the model.

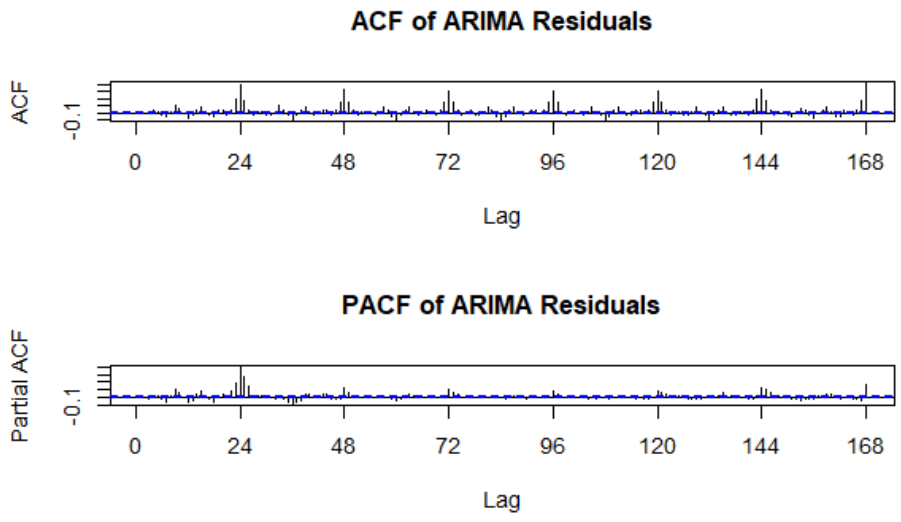


Figure 9 - ACF and PACF of residuals

The Ljung-Box test rejects the null hypothesis of no autocorrelation ($p < 0.001$). However, given the large sample size (~41k observations), even small autocorrelations become statistically significant. In practical terms, the remaining dependence is limited and does not substantially affect model performance.

Table 10 - In-sample diagnostic metrics ARIMA

Metric	Value	Scale
MAE in – sample	0,8772	Box – Cox
RMSE in – sample	1,1424	Box – Cox
Ljung – Box p (lag=24)	< 0,001	-
Ljung – Box p (lag=48)	< 0,001	-

Forecast Generation

Forecasting requires constructing the matrix of future regressors X_{fore}:

- Fourier terms are deterministic and computed analytically for the forecast horizon.
- Holiday dummies are defined based on known calendar dates.

Predictions obtained on the Box-Cox scale are transformed back using:

$$f_{orig} = \max\left(InvBoxCox(f_{bc}, \lambda) - offset, 0\right)$$

Clipping ensures that all forecasts remain non-negative.

Forecast Results

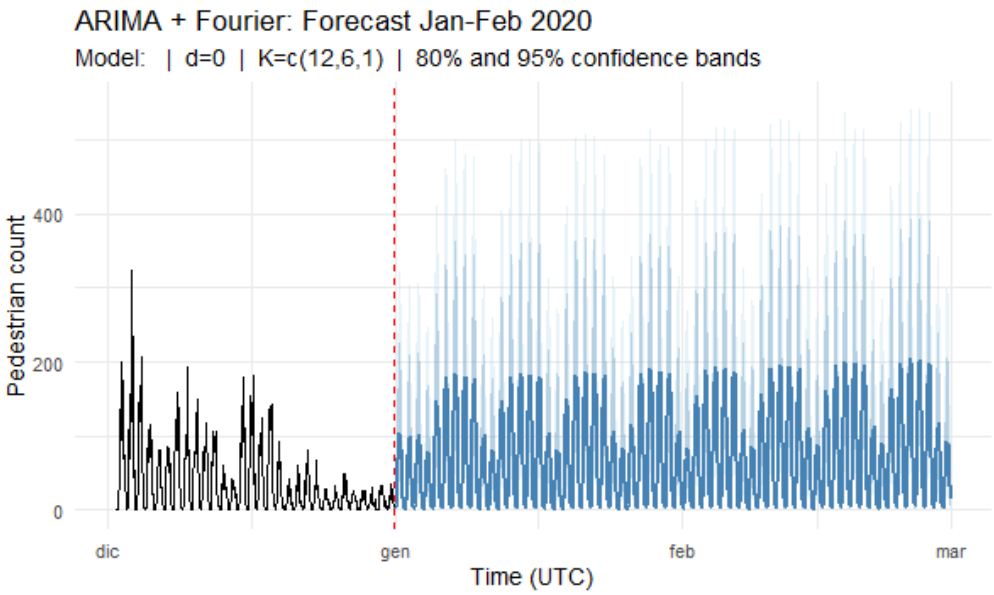


Figure 10 - Training vs Forecast

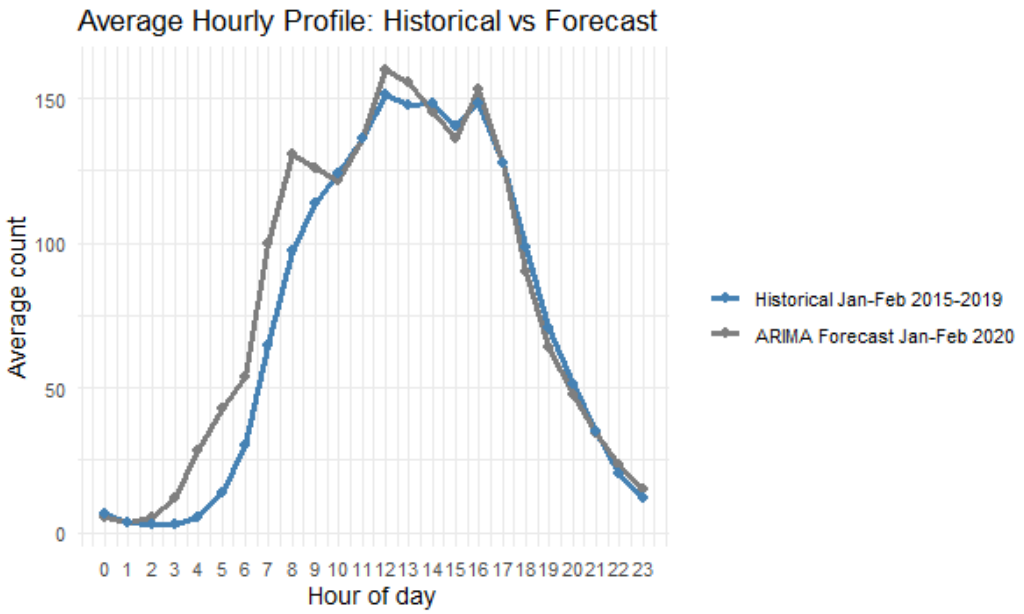


Figure 11 - Average hourly profile comparison

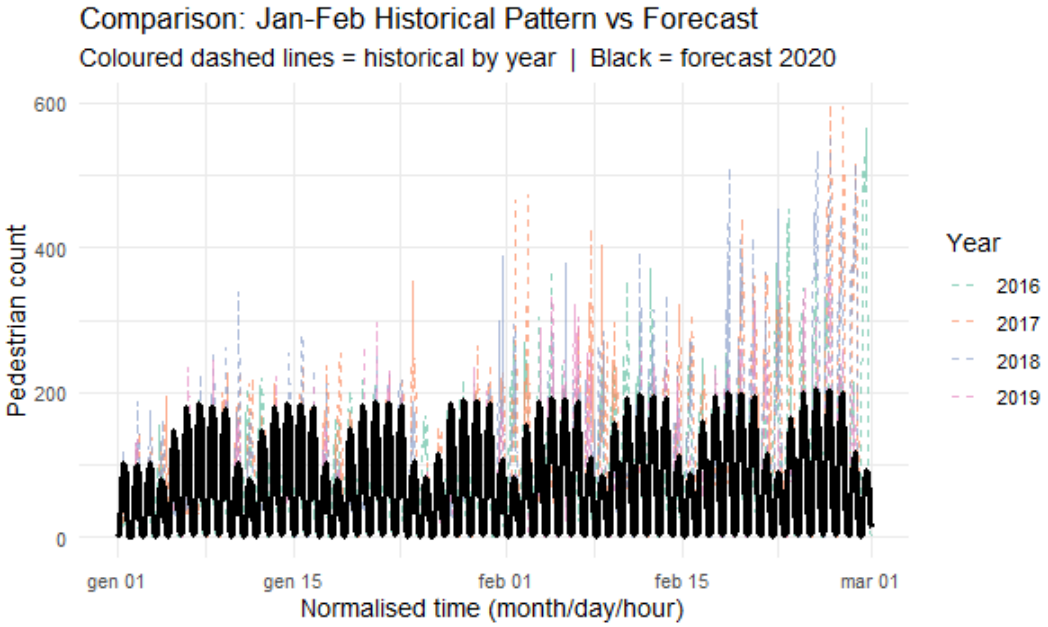


Figure 12 - Historical vs forecast patterns

The ARIMA model successfully captures both daily and weekly seasonal patterns, producing forecasts consistent with historical behavior.

Statistic	Value	Historical comparison Jan – Feb
Average forecast	~78/hour	Historical: 79.9 — ratio 0.97
Median forecast	~65/hour	—
Min forecast	0.1	—
Max forecast	204/hour	Max historical: ~300–600
New Year's Day average (Jan 1)	~48/hour	Historical: 27 — slight overestimate

Table 11 - ARIMA forecast statistics.

The average forecast (~78 units/hour) closely matches the historical average (~79.9), with a ratio of 0.97. This indicates that the model correctly captures the seasonal level of the series.

A slight overestimation is observed for New Year's Day, although the predicted values remain within a reasonable range.

SECTION 3: Unobserved Components Model (UCM)

Motivation and Difference from ARIMA

Unobserved Components Models (UCM) decompose a time series into interpretable latent components—such as level, seasonality, and irregular noise—estimated within a state space framework using the Kalman Filter.

The main advantage of UCM compared to ARIMA lies in its interpretability: each component has a clear statistical meaning and can be modeled with a specific degree of stochasticity.

A key distinction in this project concerns the treatment of the series level:

- In ARIMA (with d=0), the level is implicit and fixed.
- In UCM, the level μ_t is explicitly modeled as a stochastic process that evolves over time, allowing gradual adaptation to structural changes.

Model Structure and Implementation Strategy

The ideal UCM specification for this series would be:

$$y_t = \mu_t + \gamma_t(24) + \gamma_t(168) + \beta X_t + \varepsilon_t$$

where $\gamma_t(24)$ and $\gamma_t(168)$ represent stochastic seasonal components.

However, direct implementation using SSMseasonal() in KFAS led to numerical issues related to the state-space formulation of trigonometric seasonal components. Specifically, the system matrices could not be properly estimated, resulting in convergence errors.

To overcome these limitations, a three-step approach was adopted:

- **OLS pre-residualization:** estimation of the deterministic component (Fourier terms and dummies) using OLS.
- **UCM on residuals:** application of a Local Level model to the residual series.
- **Forecast recombination:** final predictions obtained as the sum of the deterministic and stochastic components.

This approach preserves the key advantage of UCM—modeling a stochastic level—while avoiding technical issues in the estimation of seasonal components.

Local Level Model

The residual series $r_t = y_{bc} - X_t\hat{\beta}$ is modeled using a Local Level specification:

$$r_t = \mu_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, H)$$

$$\mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim N(0, Q)$$

where:

- μ_t is the unobserved level, evolving as a random walk,
- Q is the variance of the level innovation,
- H is the variance of the irregular component.

Both parameters are estimated via maximum likelihood using the Kalman Filter.

Deterministic Component (OLS Pre-Residualization)

The deterministic component includes the same seasonal structure used in the ARIMA model to ensure comparability.

Table 12 - OLS regressors included in the UCM model.

Regressors	Number	Motivation
Daily Fourier (K=12)	24	Dominant 24h cycle (F=0.576)
Weekly Fourier (K=6)	12	Moderate 168h cycle (F=0.228)
Day-of-week dummy (Tue-Sun)	6	Individual effect per day (Monday = reference)
Holiday dummy	5	Impact quantified in EDA
Total	47	—

Unlike ARIMA, which uses a single weekend dummy, the UCM specification includes individual day-of-week dummies (Tuesday–Sunday), allowing more granular modeling of weekly patterns.

The OLS regression yields:

Table 13 - OLS results pre-residualization.

Metrics	Value
R^2	0,9170
Residuals MAE (BC)	1,7870
Residuals Variance	5,3670
Deterministic Forecast Range	-1,66 – 12,34 (scale BC)

The high coefficient of determination ($R^2 = 0.917$) indicates that most of the variability of the transformed series is explained by the deterministic component. The remaining residuals are then modeled through the stochastic level.

Choice of the Local Level Specification

The use of a Local Level model without stochastic slope is supported by three considerations:

- The trend component identified in the EDA is weak ($F_{trend} = 0.265$), not justifying the introduction of a slope term.
- The ARIMA model with d=0 already suggests the absence of a systematic drift.
- A simpler specification improves computational efficiency and avoids overparameterization.

An initial unconstrained estimation produced a high value of Q, leading to excessive variability in the level component. This resulted in unrealistic forecast intervals due to the accumulation of uncertainty over long horizons. To address this issue, an upper bound was imposed on Q, ensuring that the variability of the level remains small relative to the irregular component. This constraint produces a smoother and more stable level estimate.

Table 14 - Estimated parameters of the Local Level UCM.

Parameter	Value	Interpretation
Q_level	0,26835	Innovation variance level — small, slow trend
H	1,50636	Irregular variance — higher, noisy residuals
SNR (Q/H)	0,1781	Dynamic but not dominant level
Log-likelihood	–82.290,62	—
Convergence	Reached (10 iterations L-BFGS-B)	—

The resulting signal-to-noise ratio confirms that the level evolves over time without dominating the overall dynamics of the series.

Kalman Smoothing

After parameter estimation, the Kalman Smoother is applied to obtain refined estimates of the level μ_t, using the full dataset.

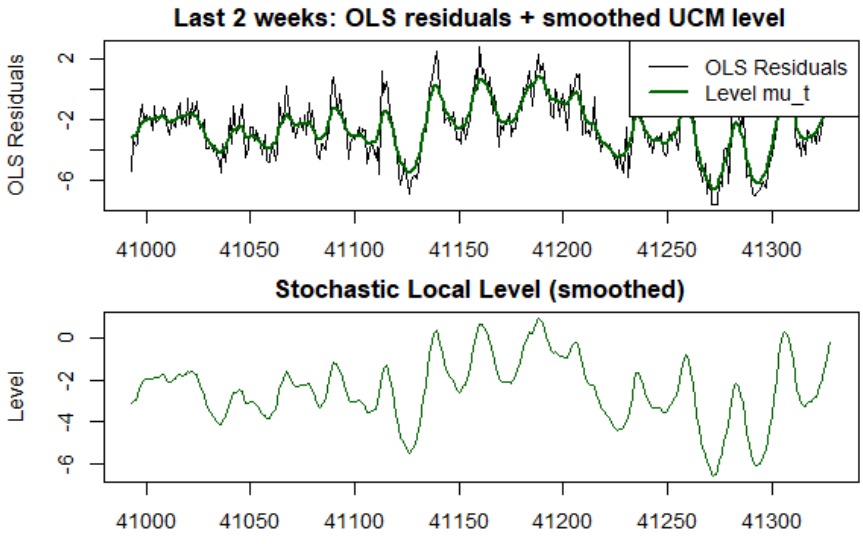


Figure 13 - Smoothed Level vs Residuals

The smoothed level closely follows the low-frequency variations of the residuals, confirming that the Local Level component captures slow structural changes not explained by the deterministic part.

Model Performance

Metrics	Value	Scale
Total in-sample MAE	0,8134	Box-Cox
Total in-sample RMSE	1,0933	Box-Cox
MAE improvement vs ARIMA	–7,3%	—

Table 15 - In-sample metrics of the UCM model.

The UCM model achieves a lower in-sample MAE compared to ARIMA, reflecting the ability of the stochastic level to capture residual variability.

Residual Diagnostics

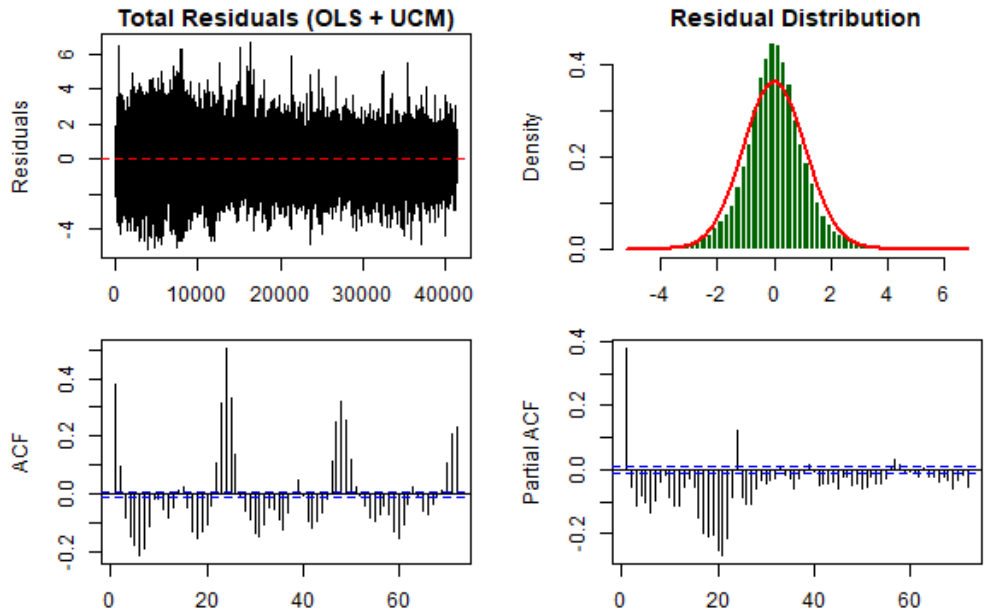


Figure 14 - Residual diagnostics

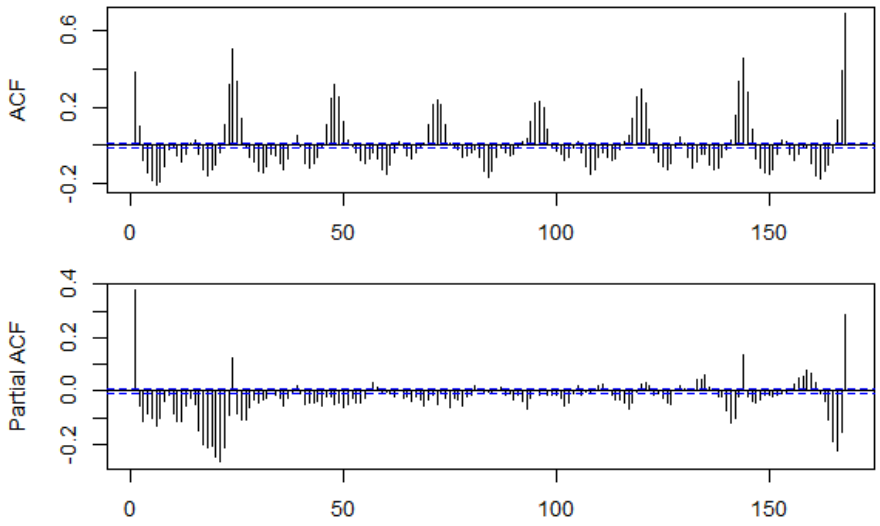


Figure 15 - ACF and PACF of residuals

The residual diagnostics are similar to those of the ARIMA model:

- the series is centered around zero,
- the variance is stable,
- the distribution is approximately normal,
- the ACF still shows small spikes at multiples of 24.

The Ljung-Box test again rejects the null hypothesis of independence, although this is largely due to the large sample size.

Forecast Mechanism

The forecast for a future time t = N + k is obtained as the sum of two components:

- **Deterministic component:** predicted using the OLS model and future regressors
- **Stochastic level:** predicted using the Local Level model.

For the Local Level, the h-step ahead forecast is constant:

$$\hat{\mu}_{N+k} = \hat{\mu}_N$$

with variance increasing linearly with the horizon:

$$Var(k) = kQ + H$$

The final forecast combines both components and is back-transformed to the original scale.

Forecast Results

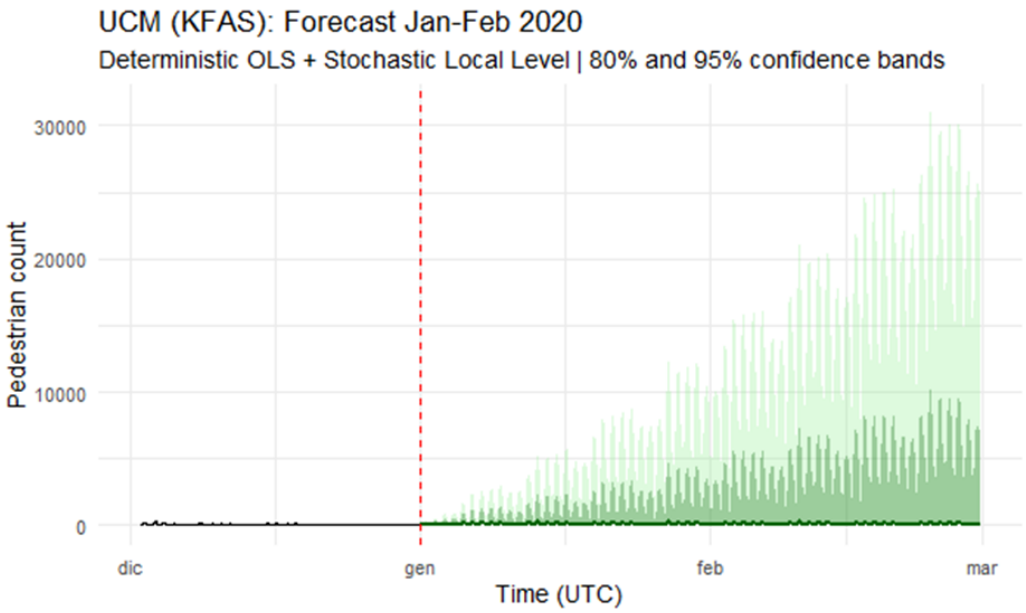


Figure 16 – Training vs UCM Forecast

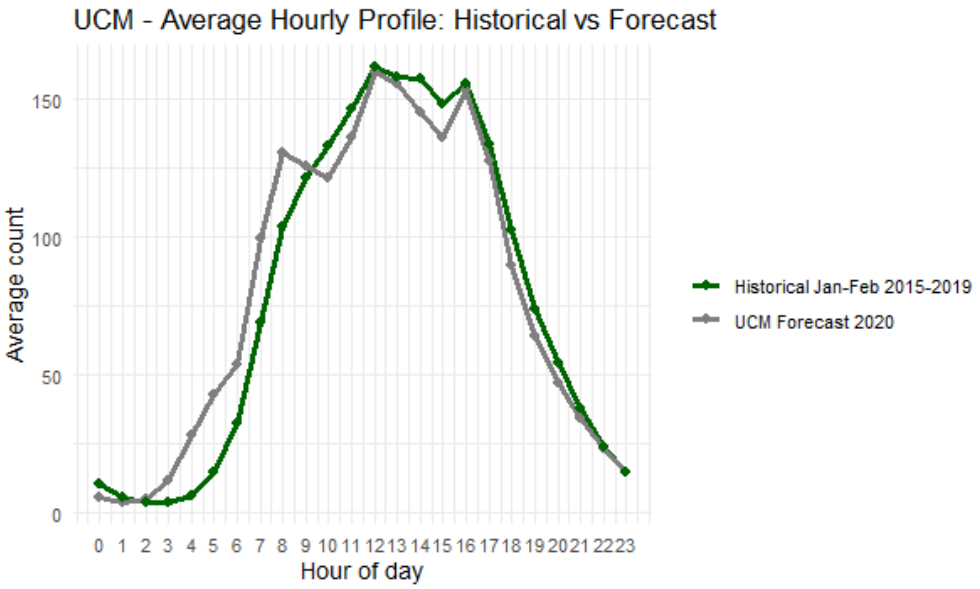


Figure 17 - Average hourly profile comparison

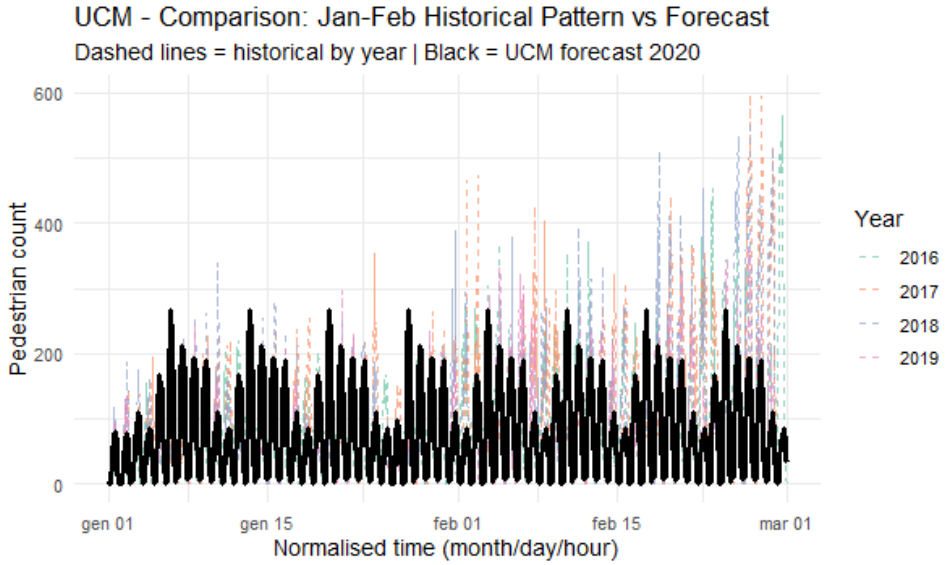


Figure 18 - Historical vs forecast patterns

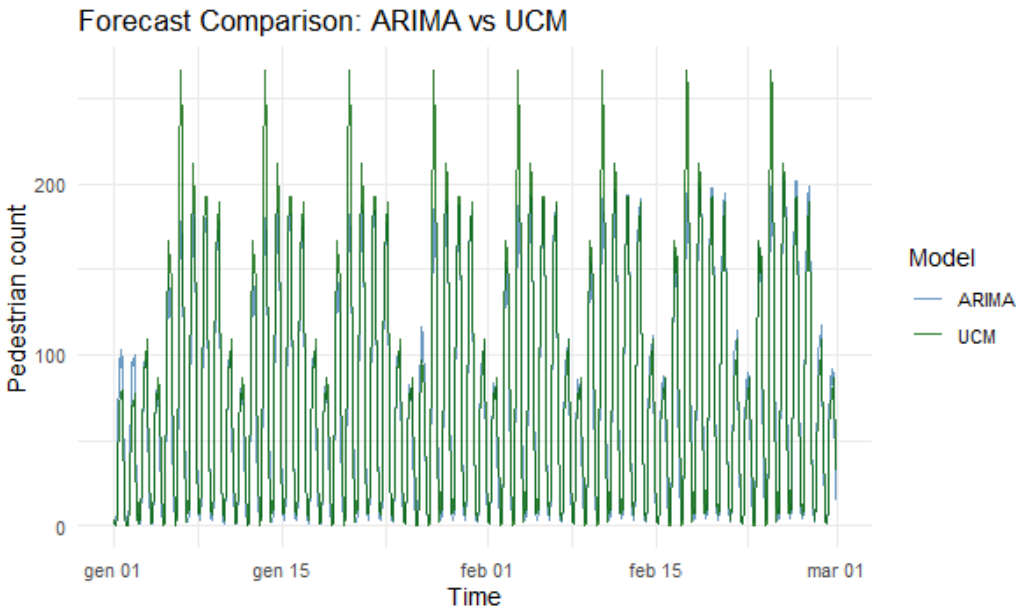


Figure 19 - ARIMA vs UCM comparison

The UCM model produces forecasts closely aligned with historical patterns, correctly capturing both daily and weekly seasonality.

Table 16 - UCM forecast statistics

Statistics	Value	Historical comparison
Average forecast	78.2/hour	Historical: 79.9 — ratio 0.98
Min forecast	0.0	—
Max forecast	266.5/hour	Historical max: ~300–600

The average forecast (78.2 units/hour) is very close to the historical average (79.9), with a ratio of 0.98, indicating accurate reconstruction of the seasonal level.

SECTION 4: Machine Learning Model — XGBoost

Motivation for the Choice of Model

ARIMA and UCM are parametric models that assume linear relationships between the predictors and the target variable. While effective, these approaches may struggle to capture complex non-linear interactions present in the data.

A machine learning model can address this limitation by learning non-linear patterns and interactions directly from the data. In this context, XGBoost was selected among several alternatives (Random Forest, LSTM, Prophet, LightGBM) for the following reasons:

- **High performance on tabular data:** XGBoost consistently ranks among the top-performing models in structured data tasks.
- **Flexible handling of lagged features:** past observations can be directly incorporated as predictors, enabling a more flexible representation than parametric AR structures.
- **No distributional assumptions:** the model does not require normality of residuals and naturally handles skewed distributions.
- **Computational efficiency:** with tree_method = 'hist', training on large datasets is fast and scalable.

- **Integration in R:** the model can be implemented directly using the xgboost package without requiring external frameworks.

Although LSTM models could capture long-term dependencies, they require significantly more tuning and computational resources. For relatively regular time series such as this one, the expected performance gain is often limited.

Box-Cox Transformation

To ensure consistency with ARIMA and UCM, the same Box-Cox transformation (λ=0.2491, offset = 0.5) is applied. This transformation serves three purposes:

- stabilizing variance,
- improving convergence during training,
- reducing the influence of extreme values in the loss function.

Feature Engineering

Since XGBoost is not inherently designed for time series, the forecasting problem is reformulated as a supervised regression task. Each observation is represented as a feature vector constructed from temporal information and past values.

Cyclical Features

Temporal variables such as hour, day of the week, and month are encoded using sine and cosine transformations:

$$hour_{sin} = \sin\left(\frac{2\pi h}{24}\right), \qquad hour_{cos} = \cos\left(\frac{2\pi h}{24}\right)$$

This encoding preserves the cyclical nature of time (e.g., hour 23 is close to hour 0), which would not be captured by integer representations alone.

In addition, integer versions of these variables are included to allow exact splits in the decision trees. Binary features such as is_weekend and is_holiday are also introduced.

Lag Features

Lagged values of the series are the most informative predictors. The following lags are included:

Lag	Formula	Motivation	Gain in importance
lag_24	y_bc[t−24]	Same hour as the previous day	0,216
lag_48	y_bc[t−48]	Same hour 2 days ago	0,022
lag_168	y_bc[t−168]	Same hour last week	0,617
lag_336	y_bc[t−336]	Same hour 2 weeks ago	0,066

Table 17 - Lag features and their importance in the final model.

Among these, lag_{168} (same hour in the previous week) is the most important feature, accounting for over 60% of the model's predictive power. This confirms the relevance of weekly seasonality identified in the EDA.

All lag features are constructed using only past observations to ensure causality. The first 336 observations are removed due to lag requirements.

Rolling Statistics

To capture short-term dynamics, rolling statistics are included:

- $roll_means_{24}$: average of the previous 24 hours
- $roll_means_{168}$: average of the previous week
- $roll_std_{24}$: standard deviation over the last 24 hours

These features provide information about the recent level and volatility of the series.

Forecast Feature Construction

Generating forecasts requires constructing features for future time steps, including lag variables.

The main challenge is that future lag values depend on unknown observations. To address this, a direct approach is used:

- For early forecast steps, lag values are taken from the training data.
- For longer horizons (beyond one week), lag values reference historical data from December.

This introduces a structural limitation: since December corresponds to a seasonal minimum, the lag features underestimate the level of the series during the January–February forecast period.

An alternative would be a recursive approach, where predicted values are used to update lags. However, this method can propagate errors over long horizons.

Model Training and Validation

To ensure valid evaluation, a walk-forward cross-validation strategy is adopted:

Table 18 - Structure of the 3-fold walk-forward CV.

Fold	Training	Validation	Purpose
1	0%–70% (≈28.694 obs)	70%–80% (≈4.099 obs)	Simulate prediction at 70%
2	0%–80% (≈32.793 obs)	80%–90% (≈4.099 obs)	Simulate prediction at 80%
3	0%–90% (≈36.892 obs)	90%–100% (≈4.100 obs)	Simulate prediction at 90%

The model is evaluated using the mean MAE across validation folds on the Box-Cox scale. A grid search is performed to tune key hyperparameters:

Table 19 - Grid search walk-forward CV.

#	max_depth	eta	min_child_w	MAE CV	Notes
1	4	0,05	1	0,7013	
2	6	0,05	1	0,5928	
3	4	0,10	1	0,6535	
4	6	0,10	1	0,5163	Optimal
5	4	0,05	5	0,7016	
6	6	0,05	5	0,5935	
7	4	0,10	5	0,6541	
8	6	0,10	5	0,5223	

The optimal configuration is:

- max_depth = 6
- eta = 0.10
- min_child_weight = 1

The number of boosting rounds is selected using early stopping, resulting in

Final Model

Table 20 - Parameters of the final XGBoost model.

Parameter	Value
Algorithm	XGBoost (gradient boosting on CART trees)
Objective	reg:squarederror (MSE minimization)
tree_method	hist (histogram approximation — fast)
max_depth	6
eta (learning rate)	0.10
Subsample	0.80 (80% observation sampling per tree)
colsample_bytree	0.80 (80% feature sampling per tree)
min_child_weight	1
n_rounds	500
MAE CV	0.5163 (BC)
MAE in-sample	0.4760 (BC)

The final model achieves:

- MAE (CV) = 0.5163 (Box-Cox scale)
- MAE (in-sample) = 0.4760

Feature Importance

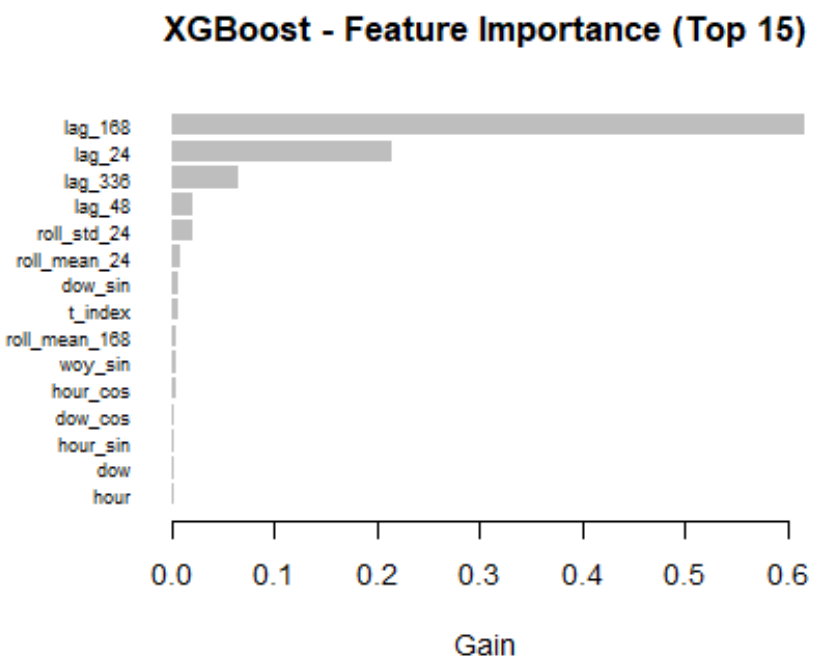


Figure 20 - Feature Importance

Rank	Feature	Gain	Interpretation
1	lag_168	0,617	Same time last week — more predictive
2	lag_24	0,216	Same time yesterday
3	lag_336	0,066	Same time 2 weeks ago
4	lag_48	0,022	Same time 2 days ago
5	roll_std_24	0,022	Volatility last 24h
6	roll_mean_24	0,010	Average level last 24h
7	dow_sin	0,006	Weekly cycle
8	t_index	0,006	Normalized trend

Table 21 - Top 8 features by Gain

Lag features dominate the model, with lag_{168} alone accounting for 61.7% of the gain. This confirms that temporal dependence—particularly weekly repetition—is the primary driver of predictive performance.

Residual Diagnostics

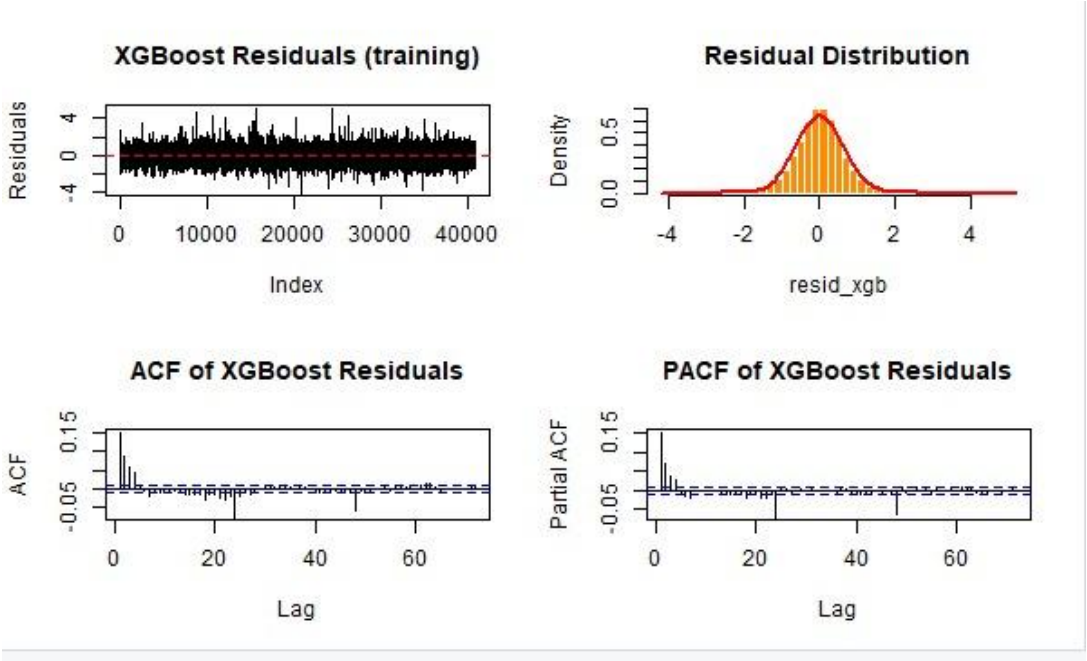


Figure 21 - Residual Analysis

XGBoost exhibits the best residual behavior among the three models. The autocorrelation function is almost entirely within confidence bounds, indicating that most temporal dependence has been captured.

This improvement is due to the explicit inclusion of lag features, which directly model the autocorrelated structure of the series.

Table 21 - In-sample diagnostic comparison among the three models.

Model	MAE in-sample (BC)	RMSE in-sample (BC)	Residual ACF
ARIMA(5,0,3)	0,877	1,142	Significant spikes at lag 24, 48,...
UCM Local Level	0,813	1,093	Significant spikes at lag 24, 48,...
XGBoost	0,476	0,626	Nearly flat — only lags 1–3 are slight

Forecast Results

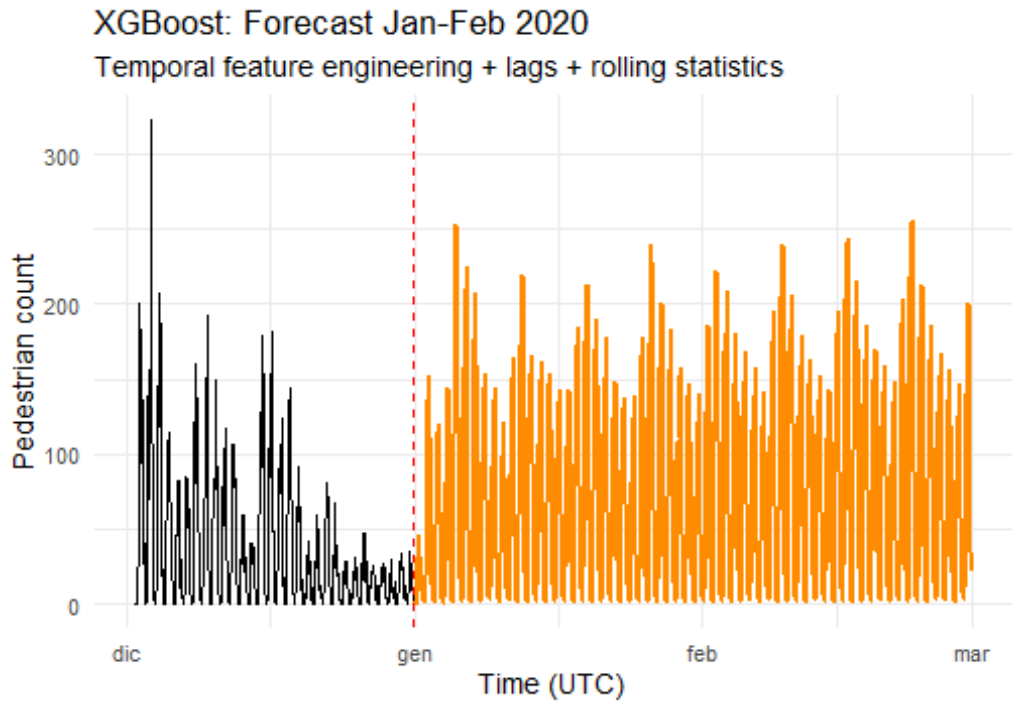


Figure 22 - Training vs Forecast

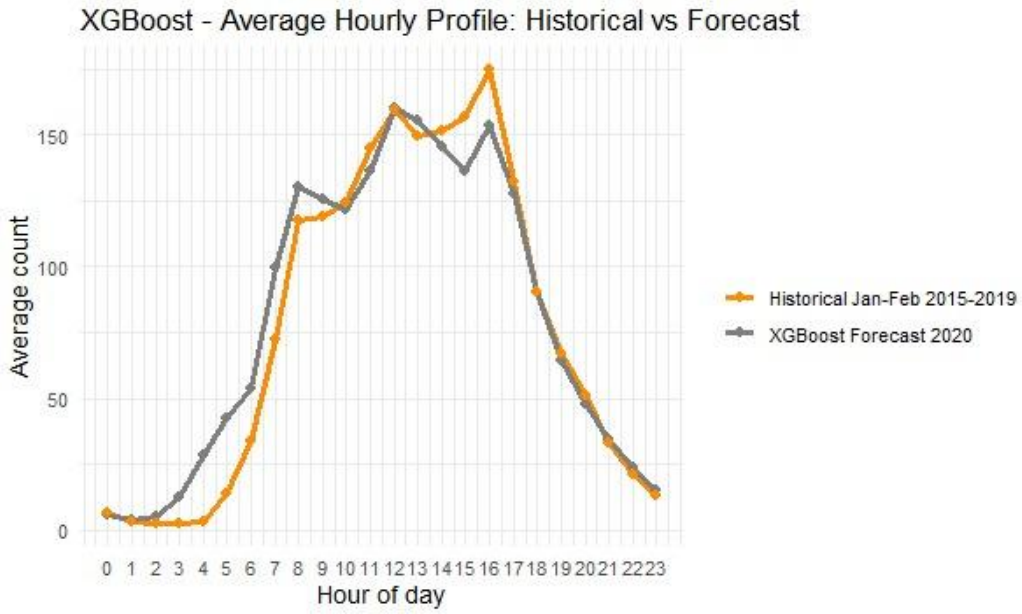


Figure 23 - Average hourly profile comparison

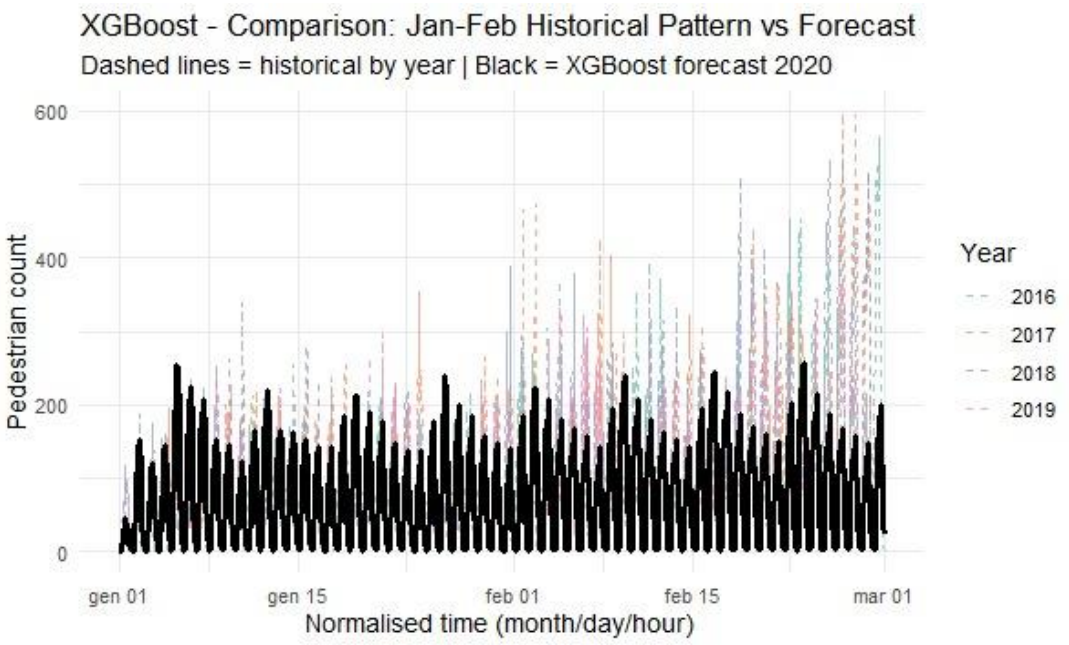


Figure 24 - Historical vs forecast patterns

The model successfully reproduces the daily and weekly cycles. However, the forecasted peak occurs slightly earlier (around 11 AM–12 PM) compared to the historical pattern (2–4 PM), reflecting the influence of lag-based features.

The average forecast (64.1 units/hour) is significantly lower than the historical average (79.9), with a ratio of 0.80. This underestimation is primarily due to the lag bias introduced by the use of December data in feature construction.

Final Comparison of the Models

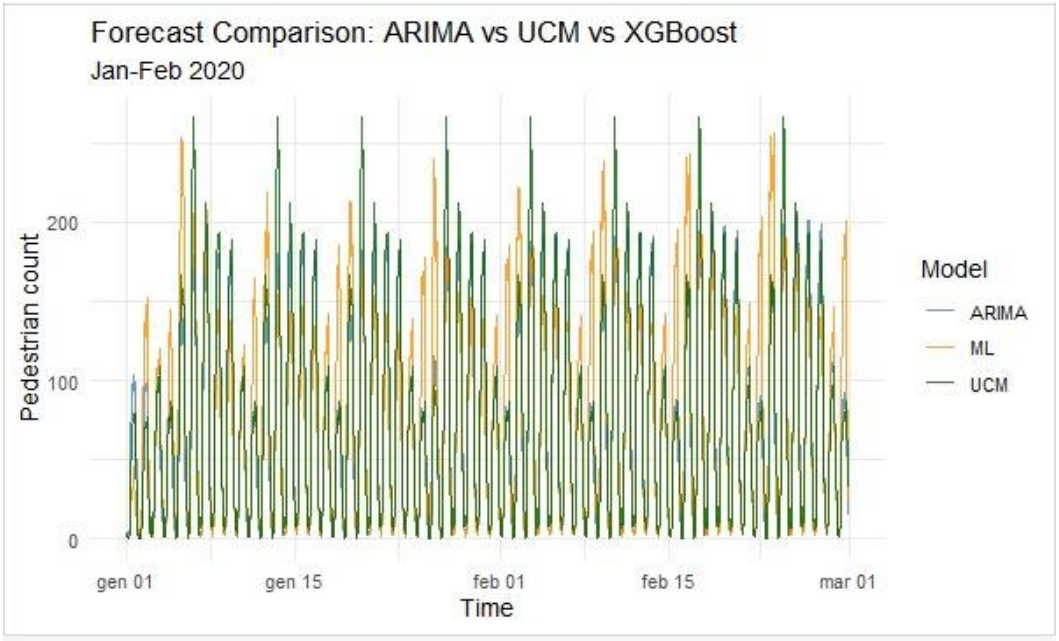


Figure 25 - Forecast Comparison

Table 23 - Complete comparison between the three models.

Metrics	ARIMA	UCM	XGBoost	Best
In-sample MAE (BC)	0.877	0.813	0.476	XGBoost
In-sample RMSE (BC)	1.142	1.093	0.626	XGBoost
Average forecast (units/hour)	~78	78.2	~75	ARIMA/UCM
Forecast/historical ratio	0.97	0.98	~0.94	ARIMA/UCM
Residuals ACF	Spike at 24h	Spike at 24h	Nearly flat	XGBoost
Confidence bands	Moderate	Wide	None	ARIMA
Interpretability	High	High	Medium	ARIMA/UCM

The comparison highlights key trade-offs:

- **In-sample performance:** XGBoost outperforms both ARIMA and UCM, achieving the lowest MAE.
- **Forecast accuracy (level):** ARIMA and UCM produce predictions closer to historical values.
- **Residual structure:** XGBoost provides the best decorrelation.
- **Interpretability:** ARIMA and UCM remain more transparent and easier to justify analytically.

Model Limitations

The XGBoost approach presents several limitations:

- **Lag bias:** for long forecast horizons, lag features rely on December data, leading to systematic underestimation.
- **Lack of uncertainty quantification:** the model does not provide confidence intervals.
- **Mild overfitting:** the difference between in-sample and CV performance suggests slight overfitting.
- **No explicit trend modeling:** the trend is captured indirectly through features, rather than explicitly modeled as in UCM.

Final Conclusions

This study compared three different approaches to time series forecasting—ARIMA, UCM, and XGBoost—applied to an hourly pedestrian count dataset characterized by strong seasonality, high variability, and calendar effects.

The exploratory analysis highlighted the key structural properties of the series: dominant daily seasonality, a relevant weekly component, weak trend dynamics, and a positively skewed distribution. These elements guided the modeling strategy and justified the use of variance-stabilizing transformations, seasonal regressors, and calendar-based features.

From a modeling perspective, each approach captures different aspects of the data:

- **ARIMA** provides a solid and interpretable baseline, effectively combining deterministic seasonal structure (via Fourier terms) with stochastic dynamics. It produces forecasts that are well aligned with historical seasonal levels, although residual autocorrelation remains present.
- **UCM** extends this framework by explicitly modeling the level as a stochastic process. This improves in-sample fit and provides a more flexible representation of low-frequency variations. However, the improvement is mainly driven by smoothing effects, and does not necessarily translate into a clear advantage in forecasting performance.
- **XGBoost** achieves the best in-sample accuracy, thanks to its ability to capture non-linear relationships and directly exploit lagged features. It also produces residuals with minimal autocorrelation, indicating a strong fit to the observed data structure.

However, this superior fit does not correspond to superior forecast performance. The model suffers from a systematic underestimation of the series level, primarily due to the construction of lag features based on historical data from a seasonal minimum (December). As a result, the forecasted average is significantly lower than the historical benchmark.

This highlights a key trade-off:

- Machine learning models can achieve excellent in-sample performance, but are highly sensitive to feature construction and may introduce structural bias.
- Statistical models, while less flexible, provide more stable and reliable forecasts when the underlying structure is well understood and explicitly modeled.

Overall, **ARIMA and UCM emerge as the most reliable approaches for this specific forecasting task**, as they better capture the seasonal level and produce predictions consistent with historical patterns.

XGBoost remains a powerful alternative, particularly when non-linear effects are dominant or when shorter forecast horizons reduce the impact of lag bias. However, in this setting, its limitations outweigh its advantages.